

Davis Baldwin

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Education

BHA

05/2026

Carnegie Mellon University | Pittsburgh, PA

- Phi Beta Kappa Inductee
- 3.78 GPA, University Honors, College Honors
- *Relevant Coursework:* Concepts of Mathematics, Fundamentals of Programming and Computer Science, Introduction to Data Structures, Principles of Computing, Reasoning with Data, Nature of Reason
- Vice President of The Originals A Cappella
- Languages: English (Fluent), Spanish (Fluent)

Work History

Research Fellow

05/2025 - 08/2025

Carnegie Mellon University | Pittsburgh, PA

- Conducted research in improvisational composition and reharmonization, investigating the acquisition of music as a language.
- Composed, performed, and produced a fully improvised, seven-movement piano suite as the centerpiece of the research.
- Independently performed all mixing and mastering engineering for the recorded suite.
- Listen: youtu.be/7EuehU9jKHM

Software Engineer Intern

05/2024 - 08/2024

Gamma Technologies LLC | TA Associates, Insight Partners portfolio company, Westmont IL

A physics-based engineering software solution serving automotive, energy, and data center industries

- Improved and optimized GUI components across the Gamma suite as a member of the GUI team.
- Debugged and troubleshot Java application issues across the Gamma technology stack — sophisticated Java applications, a custom distributed-computing solution, 3D graphics tooling, and high-performance data storage.
- Validated and edited technical documentation.

Projects

Universe — Interactive 3D Planetarium

- Built with Claude AI: a JavaScript/Three.js planetarium rendering 119,625 HYG catalog stars across four modes — sky sphere, solar system, stellar neighborhood, and procedural Milky Way galaxy.
- Sky mode includes 88 IAU constellation figures, 85 classical constellation paintings, 27 deep-sky objects, and a searchable index of 453 named stars, planets, and Messier objects.
- Solar system features accurate planetary positions computed from JPL Keplerian elements with an iterative Kepler-equation solver; includes NASA-imagery textures, true axial tilts, and Saturn's real ring-alpha texture. Animated asteroid belt (with Kirkwood gaps) and Kuiper belt.
- Galaxy mode renders a procedural Milky Way with four logarithmic spiral arms (12.5° pitch), bar/bulge, stellar halo, globular clusters, and Local Group context (Andromeda, Magellanic Clouds).
- Interactive features: drag/zoom navigation, 453-object search, data cards on click, time bar with variable playback speed (−1 to +10 yr/s), toggleable constellation/star names, magnitude limit, equatorial grid, and twinkling effect.

Sudoku Autosolver — Backtracking Puzzle Solver

- Built a Sudoku solver in Python using recursive backtracking, handling puzzles of varying difficulty.
- Implemented a built-in hint system that displays the candidate numbers possible in each square.
- Automated detection of singleton, double, and triple candidate values to narrow the search space.
- Demo: youtu.be/7QvkMWuUWXc

"Rest My Soul" — Multitracked Choral Video, set in an Infinite 3D Environment

- Composed, performed, and self-recorded all vocal parts of a one-man choir piece, then directed video recording and editing within a custom infinite 3D environment.
- Coordinated post-production with a recording engineer and mixing engineer to bring the final piece to fruition.
- Watch: youtu.be/ZdAaofDbIfM

"Life's Meaning" — Original Composition & Animated Music Video

- Composed the music and lyrics; performed all vocals and played all instruments.
- Collaborated with a mixing/mastering engineer to produce the final master.
- Partnered with an animation engineer to create a fully realized animated music video.
- Watch: youtu.be/ReZqGoQX8L8